Applications Note

mkado: a toolkit for McDonald-Kreitman tests of natural selection

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Abstract

Summary: MKADO is a Python toolkit for performing McDonald-Kreitman (MK) tests of natural selection from aligned coding sequences. It implements the standard, polarized, and asymptotic MK tests with a unified command-line interface. MKADO supports parallel batch processing of thousands of genes with near-linear scaling, and provides publication-ready visualizations including volcano plots and asymptotic alpha curves.
Availability and Implementation: MKADO is freely available at `https://github.com/andrewkern/mkado` under the MIT license. It is implemented in Python and installable via `pip`.
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1 Introduction

The McDonald-Kreitman (MK) test (McDonald and Kreitman, 1991) is among the most widely used methods for detecting natural selection at the molecular level. By comparing the ratio of nonsynonymous to synonymous changes within species (polymorphism) to that between species (divergence), the MK test provides a direct test of the neutral theory prediction that these ratios should be equal. Departures from this expectation, quantified by the neutrality index (NI; Rand and Kann, 1996) and the fraction of adaptive substitutions α (Smith and Eyre-Walker, 2002), have revealed pervasive positive selection across a wide range of taxa.

A key challenge for the standard MK test is that slightly deleterious nonsynonymous mutations, which contribute to polymorphism but rarely fix, inflate Pn relative to Dn and bias α downward (Charlesworth *et al.*, 1993; Eyre-Walker *et al.*, 2006). The asymptotic MK test addresses this by estimating α as a function of derived allele frequency and extrapolating to fixation, effectively removing the contribution of weakly deleterious variants (Messer and Petrov, 2013; Haller and Messer, 2017). The polarized MK test uses a second outgroup to assign fixed differences to specific lineages, allowing lineage-specific inference of selection (Smith and Eyre-Walker, 2002).

Despite the centrality of MK-based methods to molecular evolution, existing software implementations remain fragmented. DnaSP (Librado and Rozas, 2009) provides the standard test through a Windows-only graphical interface. The iMKT R package and web server (Murga-Moreno *et al.*, 2019) implements several extensions but requires pre-computed summary statistics and is tightly coupled to specific databases. The asymptoticMK web tool (Haller and Messer, 2017) implements the asymptotic test but does not accept raw sequence data. No existing tool provides a unified command-line interface that takes aligned coding sequences as input and supports the standard, polarized, and asymptotic MK tests together with batch processing across thousands of genes (Table 1).

Table 1. Feature comparison of MK test software. Checkmarks indicate built-in support; dashes indicate the feature is absent.

Feature	DnaSP	iMKT	asymptotic MK	MKADO
Sequence input	✓	web only	—	✓
Standard MK	✓	✓	—	✓
Polarized MK	—	—	—	✓
Asymptotic MK	—	✓	✓	✓
CLI	—	—	—	✓
Batch processing	—	—	—	✓
Parallel execution	—	—	—	✓
Multiple testing corr.	—	—	—	✓
Visualization	limited	✓	✓	✓
Open source	—	✓	✓	✓
Cross-platform	—	✓	✓	✓

Here we present MKADO, a Python package that fills this gap. MKADO takes codon-aligned FASTA files as input and implements the standard, polarized, and asymptotic MK tests with a consistent command-line interface. It supports parallel batch processing of genome-scale datasets, automatic multiple testing correction, and built-in visualization including volcano plots and asymptotic α curves. MKADO is installable via `pip` and is designed to integrate naturally with modern Python-based bioinformatics workflows.

2 Features and Implementation

MKADO is implemented in Python and provides a command-line interface built on Typer. It accepts codon-aligned FASTA files as input, requiring no pre-processing or external format conversion. Sequences are automatically partitioned into ingroup and outgroup sets by name pattern matching, supporting both multi-species alignments in a single file and separate ingroup/outgroup files. Full documentation, including tutorials and API reference, is available at `https://mkado.readthedocs.io`.

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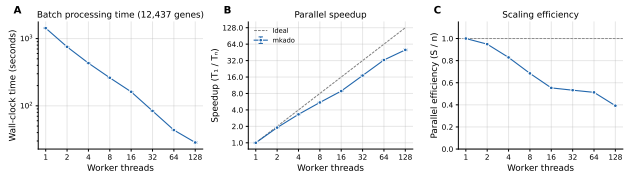

Fig. 1. Parallel scaling of MKADO batch processing on 12,437 human protein-coding genes. (**A**) Wall-clock time versus worker threads. (**B**) Speedup relative to single-threaded execution, with ideal linear scaling shown as a dashed line. (**C**) Parallel efficiency (speedup divided by thread count).

Analysis methods. MKADO implements three MK test variants. The *standard* test computes the classical 2×2 table of nonsynonymous and synonymous changes in polymorphism and divergence (Dn, Ds, Pn, Ps), with significance assessed by Fisher's exact test. Summary statistics include the neutrality index (Rand and Kann, 1996), α (Smith and Eyre-Walker, 2002), and the Direction of Selection statistic (DoS; Stoletzki and Eyre-Walker, 2011). The *polarized* test uses a second outgroup to infer the ancestral state at each fixed difference, assigning divergence to the ingroup or outgroup lineage for lineage-specific inference. The *asymptotic* test bins polymorphisms by derived allele frequency (DAF), computes α within each bin, and fits an exponential model $\alpha(x) = a + be^{-cx}$ to extrapolate the asymptotic α at $x = 1$, correcting for the contribution of weakly deleterious variants (Messer and Petrov, 2013). Confidence intervals are obtained by bootstrap resampling.

Codon change classification. When comparing codons that differ at multiple positions, the assignment of changes as synonymous or nonsynonymous is not straightforward because the order of substitutions is unknown. MKADO pre-computes all shortest mutational paths between every pair of codons and weights the synonymous and nonsynonymous contributions equally across paths, following the approach of Nei and Gojobori (1986).

Batch processing and parallelism. The `batch` command processes directories of alignment files in parallel using Python's multiprocessing facilities with configurable worker counts. Benjamini-Hochberg adjusted p-values are computed automatically across all genes. To evaluate scaling performance, we benchmarked MKADO on 12,437 human protein-coding genes aligned with chimpanzee and orangutan orthologs from OrthoMaM (Allio *et al.*, 2024), with polymorphism data from 178 Yoruba individuals in the 1000 Genomes Project (1000 Genomes Project Consortium, 2015). Processing all genes with 128 worker threads completed in 28 seconds (50× speedup over single-threaded execution), with near-linear scaling up to 32 threads and sustained efficiency at higher thread counts (Fig. 1).

Visualization. MKADO generates publication-ready plots directly from the command line. Volcano plots display $-\log_{10}(\text{NI})$ against $-\log_{10}(p)$ for each gene, with Bonferroni and nominal significance thresholds (Fig. 2). The asymptotic test produces $\alpha(x)$ curves showing the fitted model and bootstrap confidence intervals.

3 Application

To demonstrate MKADO at genome scale, we assembled a dataset of 12,437 human protein-coding genes with both polymorphism and divergence data. Codon-aligned orthologous sequences for human (*Homo sapiens*), chimpanzee (*Pan troglodytes*), and orangutan (*Pongo abelii*) were obtained from OrthoMaM v12 (Allio *et al.*, 2024). Human intraspecific polymorphism was drawn from 178 Yoruba (YRI) individuals in the 1000 Genomes Project NYGC high-coverage dataset

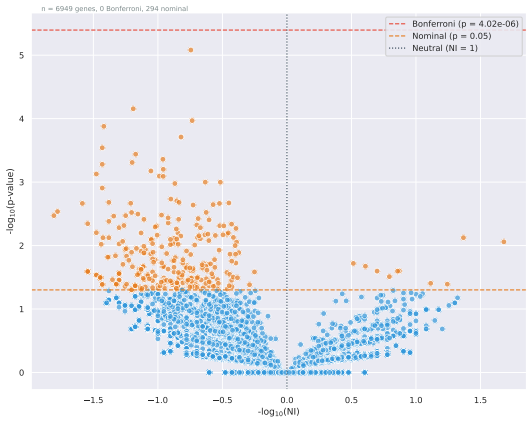

Fig. 2. Volcano plot from standard MK tests on 12,437 genes using *Pan troglodytes* as the outgroup. Orange points are nominally significant ($p < 0.05$); dashed lines indicate Bonferroni and nominal thresholds. The leftward shift of the cloud (NI > 1) reflects the well-known excess of weakly deleterious nonsynonymous polymorphism in humans.

(1000 Genomes Project Consortium, 2015), yielding 356 phased haplotypes per gene. Variants were mapped onto the OrthoMaM alignment coordinates to produce multi-species FASTA files suitable for direct input to MKADO.

Using chimpanzee as the outgroup, genome-wide pooled counts were Dn = 37,120, Ds = 65,228, Pn = 44,850, and Ps = 47,838. The per-gene standard MK test identified 465 genes with nominally significant departures from neutrality ($p < 0.05$), though none survived Benjamini-Hochberg correction, consistent with the limited power of single-gene tests at human-chimpanzee divergence levels. As expected, the majority of testable genes showed NI > 1 (Fig. 2), reflecting the well-characterized excess of slightly deleterious nonsynonymous polymorphism in humans (Eyre-Walker *et al.*, 2006). The aggregated asymptotic MK test estimated $\hat{\alpha} = 0.075$ (95% CI: -1.22 to 0.11), with the wide confidence interval reflecting uncertainty in extrapolating the frequency-dependent $\alpha(x)$ curve from the relatively shallow human-chimpanzee divergence.

Switching to orangutan as the outgroup substantially improved precision. The longer divergence time yielded roughly 2.5-fold more fixed differences (Dn = 94,578, Ds = 188,325), and the asymptotic estimate tightened to $\hat{\alpha} = 0.163$ (95% CI: 0.075 to 0.253), indicating that approximately 16% of amino acid substitutions on the human lineage were driven by positive selection. This estimate is consistent with previous analyses of human coding sequences (Messer and Petrov, 2013; Eyre-Walker *et al.*, 2006).

The entire analysis—standard MK tests, asymptotic estimation, and volcano plot generation for all 12,437 genes—completed in under one minute on a multi-core workstation using MKADO's parallel batch mode.

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